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Quantum supremacy using a programmable superconducting processor

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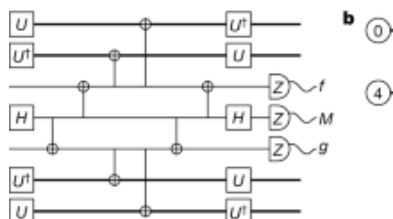
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Abstract

The promise of quantum computers is that certain computational tasks might be executed exponentially faster on a quantum processor than on a classical processor¹. A fundamental challenge is to build a high-fidelity processor capable of running quantum algorithms in an exponentially large computational space. Here we report the use of a processor with programmable superconducting qubits^{2,3,4,5,6,7} to create quantum states on 53 qubits, corresponding to a computational state-space of dimension 2^{53} (about 10^{16}). Measurements from repeated experiments sample the resulting probability distribution, which we verify using classical simulations. Our Sycamore processor takes about 200 seconds to sample one instance of a quantum circuit a million times—our benchmarks currently indicate that the equivalent task for a state-of-the-art classical supercomputer would take approximately 10,000 years. This dramatic increase in speed compared to all known classical algorithms is an

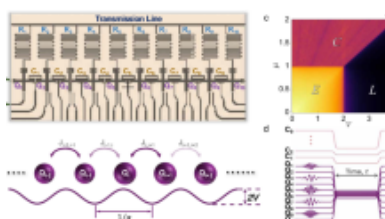
experimental realization of quantum supremacy^{[8,9,10,11,12,13,14](#)} for this specific computational task, heralding a much-anticipated computing paradigm.

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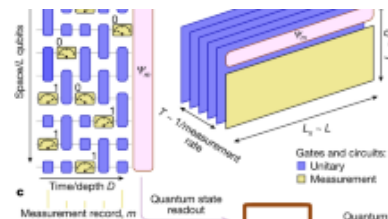
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Main

In the early 1980s, Richard Feynman proposed that a quantum computer would be an effective tool with which to solve problems in physics and chemistry, given that it is exponentially costly to simulate large quantum systems with classical computers^{[1](#)}. Realizing Feynman's vision poses substantial experimental and theoretical challenges. First, can a quantum system be engineered to perform a computation in a large enough computational (Hilbert) space and with a low enough error rate to provide a quantum speedup? Second, can we formulate a problem that is hard for a classical computer but easy for a quantum computer? By computing such a benchmark task on our superconducting qubit processor, we tackle both questions. Our experiment achieves quantum supremacy, a milestone on the path to full-scale quantum computing^{[8,9,10,11,12,13,14](#)}.

In reaching this milestone, we show that quantum speedup is achievable in a real-world system and is not precluded by any hidden physical laws. Quantum supremacy also heralds the era of noisy intermediate-scale quantum (NISQ) technologies¹⁵. The benchmark task we demonstrate has an immediate application in generating certifiable random numbers (S. Aaronson, manuscript in preparation); other initial uses for this new computational capability may include optimization^{16,17}, machine learning^{18,19,20,21}, materials science and chemistry^{22,23,24}. However, realizing the full promise of quantum computing (using Shor's algorithm for factoring, for example) still requires technical leaps to engineer fault-tolerant logical qubits^{25,26,27,28,29}.

To achieve quantum supremacy, we made a number of technical advances which also pave the way towards error correction. We developed fast, high-fidelity gates that can be executed simultaneously across a two-dimensional qubit array. We calibrated and benchmarked the processor at both the component and system level using a powerful new tool: cross-entropy benchmarking¹¹. Finally, we used component-level fidelities to accurately predict the performance of the whole system, further showing that quantum information behaves as expected when scaling to large systems.

A suitable computational task

To demonstrate quantum supremacy, we compare our quantum processor against state-of-the-art classical computers in the task of sampling the output of a pseudo-random quantum circuit^{11,13,14}. Random circuits are a suitable choice for benchmarking because they do not possess structure and therefore allow for limited guarantees of computational hardness^{10,11,12}. We design the circuits to entangle a set of quantum bits (qubits) by repeated application of single-qubit and two-qubit logical operations. Sampling the quantum circuit's output produces a set of bitstrings, for example {0000101, 1011100, ...}. Owing to quantum interference, the probability distribution of the bitstrings resembles a speckled intensity pattern produced by light interference in laser scatter, such that some bitstrings are much more likely to

occur than others. Classically computing this probability distribution becomes exponentially more difficult as the number of qubits (width) and number of gate cycles (depth) grow.

We verify that the quantum processor is working properly using a method called cross-entropy benchmarking^{[11,12,14](#)}, which compares how often each bitstring is observed experimentally with its corresponding ideal probability computed via simulation on a classical computer. For a given circuit, we collect the measured bitstrings $\{x_i\}$ and compute the linear cross-entropy benchmarking fidelity^{[11,13,14](#)} (see also [Supplementary Information](#)), which is the mean of the simulated probabilities of the bitstrings we measured:

$$\mathcal{F}_{\text{XEB}} = 2^n \langle P(x_i) \rangle_i - 1$$

(1)

where n is the number of qubits, $P(x_i)$ is the probability of bitstring x_i computed for the ideal quantum circuit, and the average is over the observed bitstrings. Intuitively, $\mathcal{F}_{\text{XEB}}$ is correlated with how often we sample high-probability bitstrings. When there are no errors in the quantum circuit, the distribution of probabilities is exponential (see [Supplementary Information](#)), and sampling from this distribution will produce $\mathcal{F}_{\text{XEB}} = 1$. On the other hand, sampling from the uniform distribution will give $\langle P(x_i) \rangle_i = 1/2^n$ and produce $\mathcal{F}_{\text{XEB}} = 0$. Values of $\mathcal{F}_{\text{XEB}}$ between 0 and 1 correspond to the probability that no error has occurred while running the circuit. The probabilities $P(x_i)$ must be obtained from classically simulating the quantum circuit, and thus computing $\mathcal{F}_{\text{XEB}}$ is intractable in the regime of quantum supremacy. However, with certain circuit simplifications, we can obtain quantitative fidelity estimates of a fully operating processor running wide and deep quantum circuits.

Our goal is to achieve a high enough $\mathcal{F}_{\text{XEB}}$ for a circuit with sufficient width and